

Welcome and thank you for listening to us today. In this presentation, we will share insights from our project focused on forecasting Austria's electricity load for the upcoming day.

Austria's energy market has been fully liberalized since 2001 (for electricity) and 2002 (for gas), characterized by the unbundling of grid operations and energy supply. While grid infrastructure remains under regulated monopoly control by the government, energy suppliers operate in a competitive market. A defining feature of the Austrian power sector is its high share of renewable energy – particularly hydropower – which accounts for the majority of domestic electricity generation.

However, the strong dependence on weather conditions leads to significant volatility in electricity supply. Combined with the increasing decentralization and complexity of the energy system, this creates growing challenges for short-term system balancing. Accurate load forecasting – especially on a quarter-hour basis – is therefore essential for grid operators, suppliers, and market participants to ensure supply security, cost-efficiency, and operational stability.

Forecasting models play a key role in this context. They are used to predict electricity demand, optimize generation, manage market risks, and support bidding strategies in day-ahead and intra-day markets. These models typically rely on historical load data, time series features, and exogenous variables such as weather or calendar effects. We used the given SQL Database with load and data for energy generation as well as external weather data from Open Meteo via a JSON API.

In our Energy Load Prediction Model, we incorporate engineered features to better capture temperature-dependent effects on demand. For example, heating demand is modeled as a nonlinear function that increases more strongly when temperatures drop below 15 °C, and cooling demand similarly increases above 22 °C, both raised to a power greater than one to emphasize extreme conditions. We used a variable as Exponent to Experiment with the gradient of the function. This approach helps to more accurately represent consumption patterns driven by weather extremes. Additionally, we use lag features derived from past load values and other exogenous variables. We cleansed all the Data by discarding non functional rows(not all attributs filled out) as well as transforming calendar data to actual features like Weekday or Holiday. We also used rolling metrics as well as day and year cyclus representing features.

Our modeling framework combines a baseline linear regression with a more advanced machine learning algorithm, XGBoost (Extreme Gradient Boosting). Linear regression provides a simple, interpretable model for capturing consistent, linear relationships between input variables and load. XGBoost, on the other hand, is an ensemble method that sequentially builds decision trees, enabling it to model complex, nonlinear dependencies such as interactions between weather, time, and historical load. Regularization and tree pruning techniques in XGBoost reduce overfitting and improve generalization on unseen data. We also hyperparameter tuned our XGBoost model to get the best results.

By integrating these two approaches, our model balances interpretability and predictive power. This two model approach compares a standard statistical and a more optimized Machine Learning model. Ultimately, this supports grid operators, energy suppliers, and market participants in optimizing operational decisions, enhancing supply security, and reducing balancing costs within Austria's evolving energy market."